
RS Quark Sector: Literature Reproductions and Custodial

Minimal (non-custodial) Randall-Sundrum validated against the published figures, plus what custodial symmetry changes. June 2026

Everything is the **anarchic baseline (lane A)**, the literature strawman used to validate the code. It is *not* production: production as run (lane B, fit-aligned MFV) is a ε_K wall near **7 TeV**, the FPR ideal (lane C, V_{5KM}) near **2 TeV**. So the ~ 30 TeV anarchic ε_K cloud is the strawman, not “our floor.”

1 Floor summary

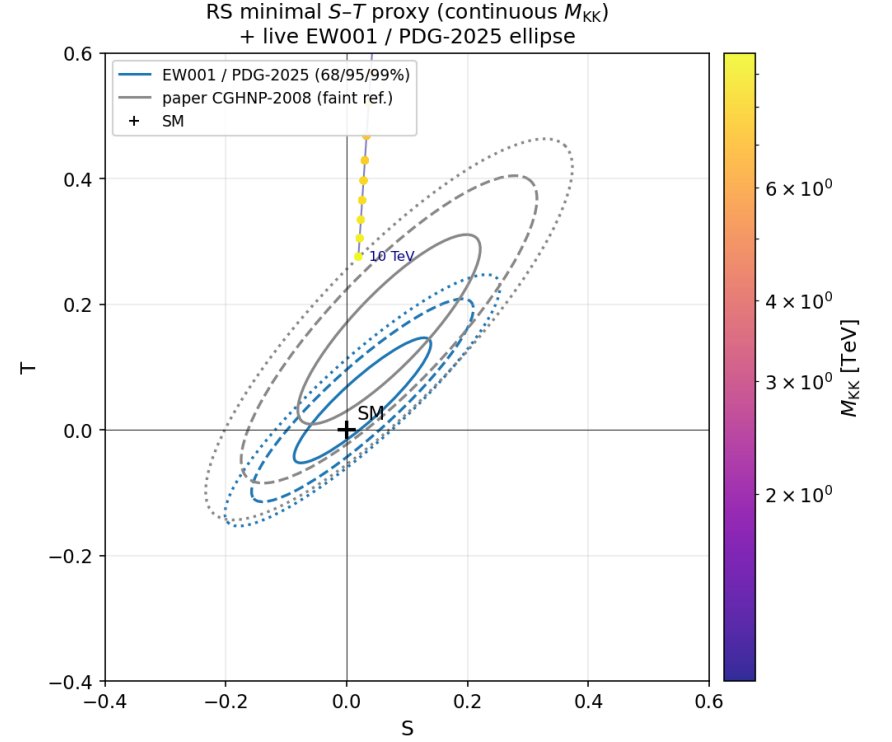
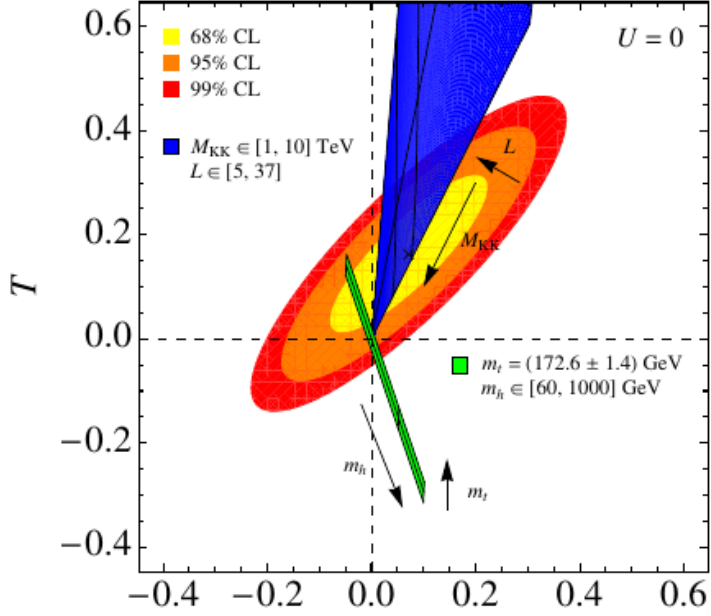
Two floors. **Typical** (median anarchic point allowed): ~ 30 TeV, set by ε_K . **Existence** (best fine-tuned point allowed): $\sim 18\text{--}20$ TeV, set by S, T, U . Flavor is tunable (align $\text{Im } M_{12} \rightarrow 0$); S, T, U is not, which is why custodial exists.

Constraint	Typical (median)	Existence (best/tuned)	Tunable?
ε_K	~ 30 TeV (lane A; B ~ 7 , C ~ 2)	< 1 TeV	yes
$\Delta m_d, \Delta m_s$	few TeV	< 1 TeV	yes
$D^0, \Delta m_K$	low	$< 1\text{--}3$ TeV	yes
$Z \rightarrow b\bar{b}$	~ 5 TeV	~ 4.6 TeV	no (gauge; c_{Q_3} set by m_t)
S, T, U	~ 20 TeV	$\sim 18\text{--}20$ TeV	no (no Yukawa freedom)
Combined	~ 30 TeV (ε_K)	$\sim 18\text{--}20$ TeV (S, T, U)	

2 Reproductions (paper left, ours right)

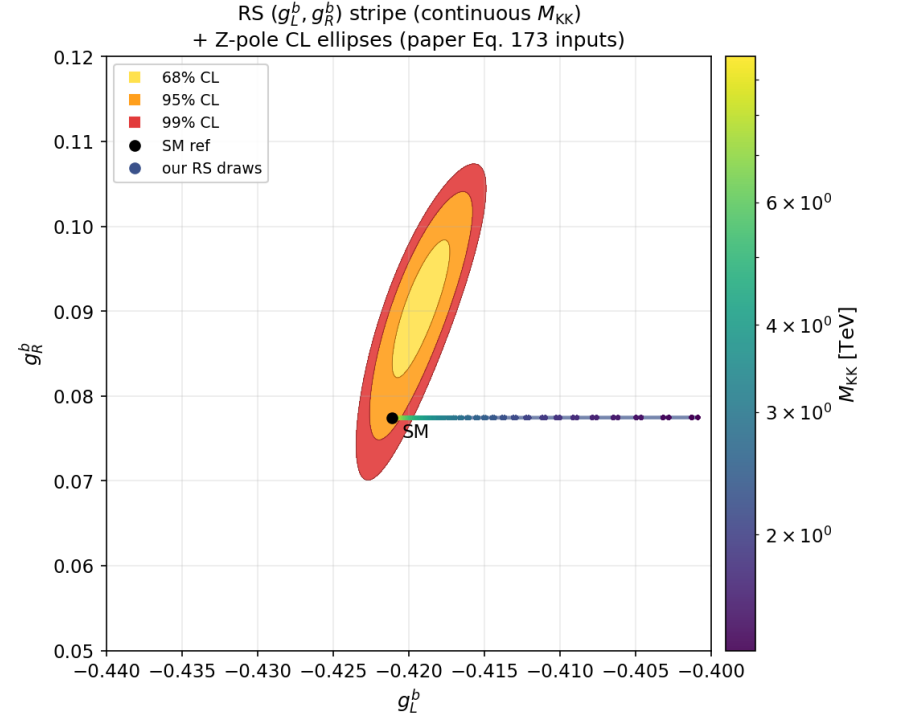
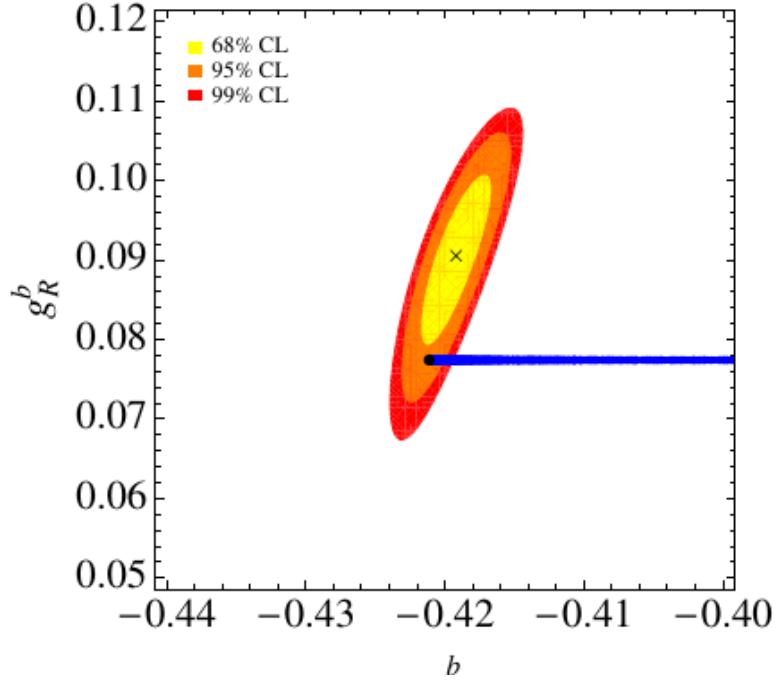
Each is the correctly cropped single minimal / S1 paper panel on the left, our fixed-code reproduction on the right, in the same axes.

2.1 S, T, U oblique (Casagrande et al. 0807.4937, Fig. 4)



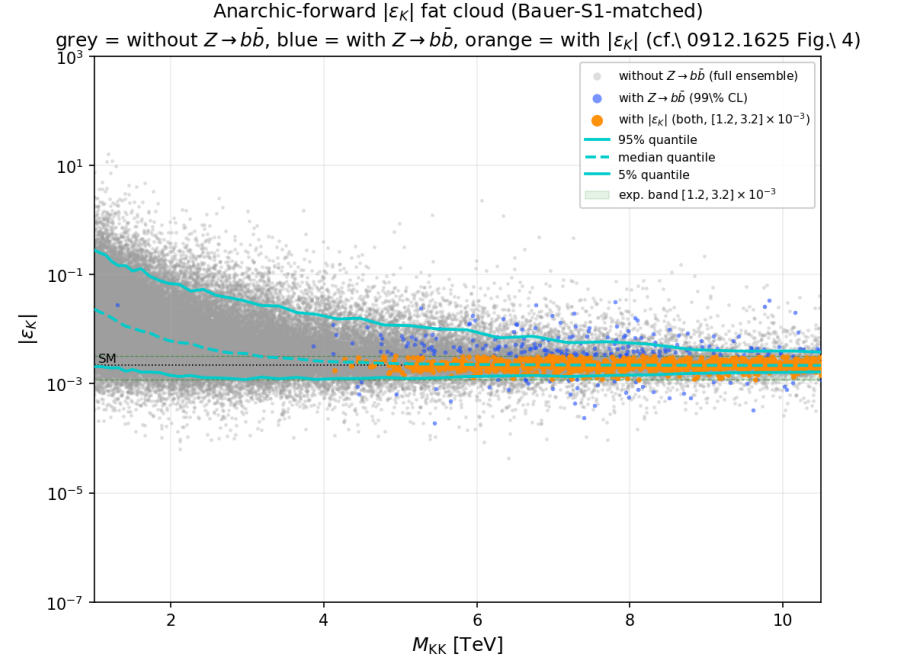
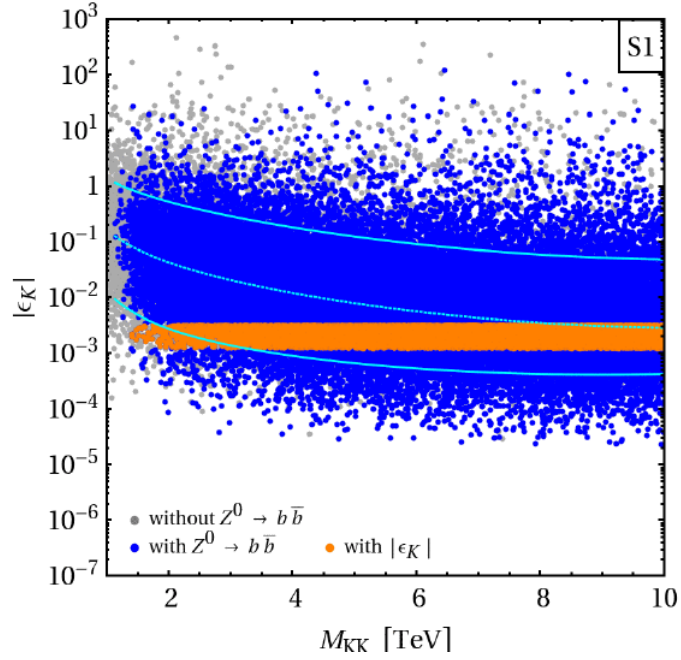
The minimal wedge climbs steeply in T (the volume-enhanced T problem) and leaves the ellipse fast. This irreducible bound sets the ~ 18 – 20 TeV existence floor.

2.2 $Z \rightarrow b\bar{b}$ couplings (Casagrande et al. 0807.4937, Fig. 8)



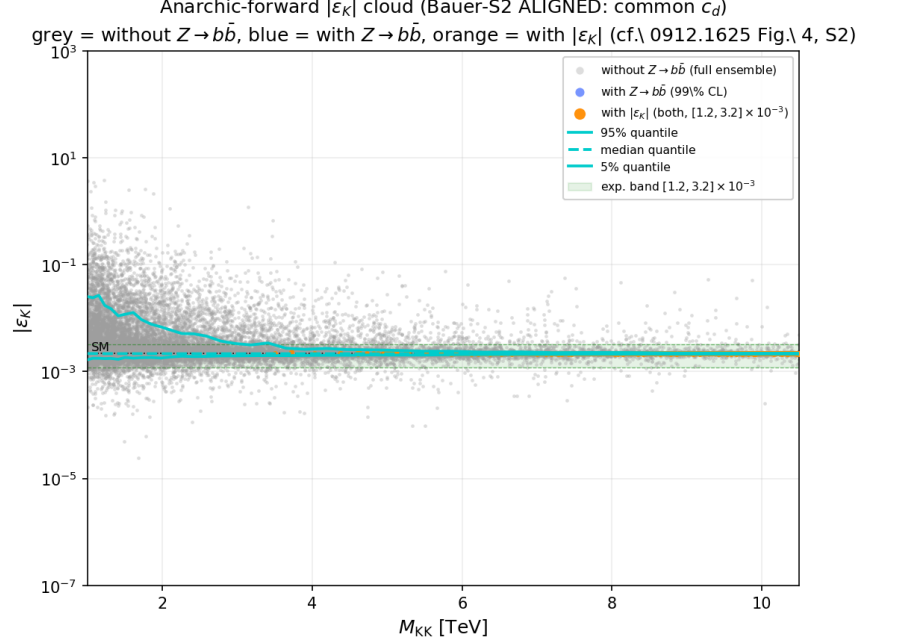
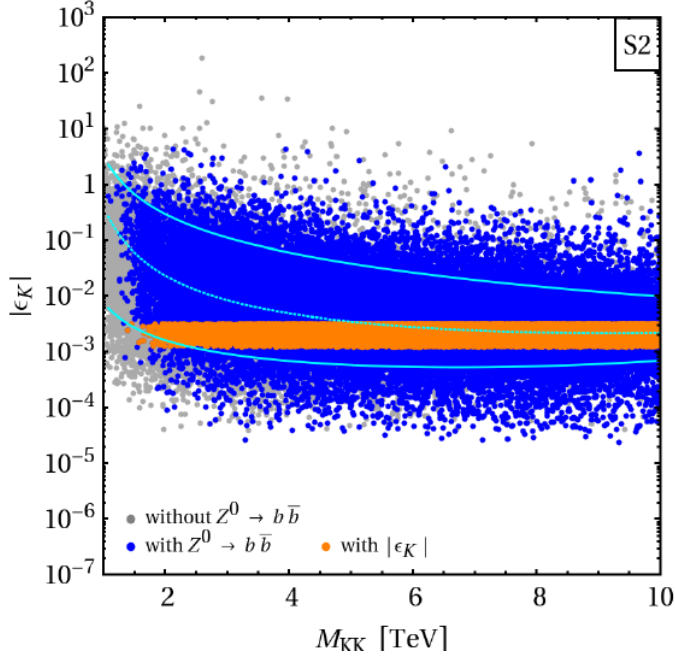
The (g_L^b, g_R^b) stripe slides to less-negative g_L^b as M_{KK} drops, staying near the SM. Gauge-dominated and weak; the bound reaches only ~ 5 TeV (post-fix).

2.3 ε_K , S1 standard (Bauer et al. 0912.1625, Fig. 4 top-left)



Anarchic $|\varepsilon_K|$ cloud, ~ 6 decades (grey = mass+CKM gated; blue = $Z \rightarrow b\bar{b}$ -consistent; orange = in the $|\varepsilon_K|$ window). **S1 “standard” = independent per-generation RH-down bulk masses** ($c_{d_1}, c_{d_2}, c_{d_3}$ each fit to masses and CKM, full anarchy). This is the scenario our production catalog implements. The 95% quantile enters the band near 10 TeV.

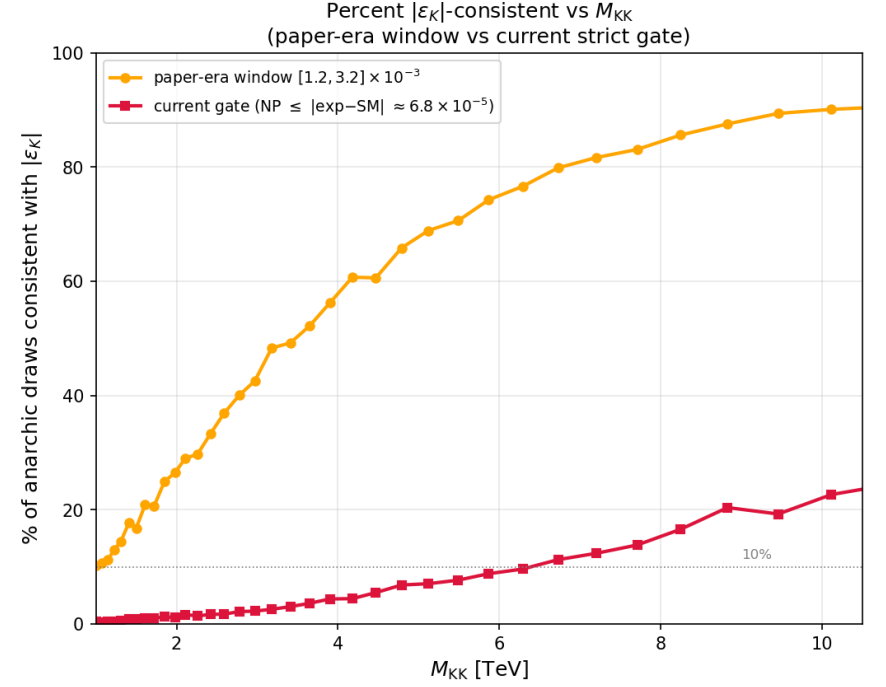
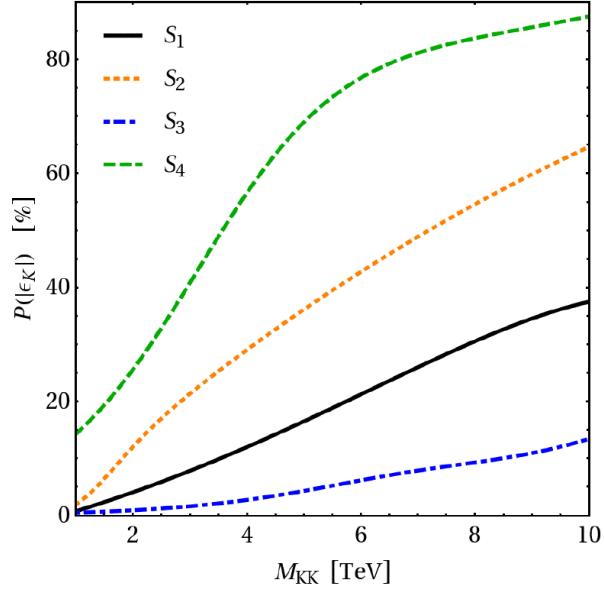
2.4 ε_K , S2 aligned (Bauer et al. 0912.1625, Fig. 4 top-right)



S1 vs S2 is exactly one change. S2 “aligned” imposes a *common* RH-down bulk mass $c_{d_1} = c_{d_2} = c_{d_3}$ (a $U(3)$ flavor symmetry on the RH down quarks); everything else (same draws, same gate, same colouring) is identical. In code the switch is a single flag (`common_cd`): both fit the three RH-down bulk masses to masses and CKM, then S2 averages them to their mean while S1 keeps them independent. That single alignment suppresses the down-sector $\Delta F = 2$ mixing, so the ε_K cloud collapses toward the band (floor ~ 10 TeV in S1 $\rightarrow \sim 2\text{--}3$ TeV in S2) and the orange in-band fraction fattens. It is a different model (an extra flavor symmetry), not our scan, and not custodial.

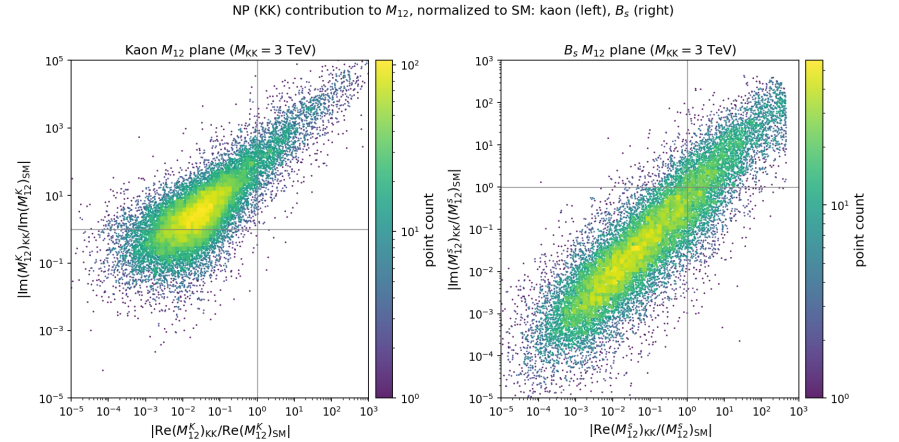
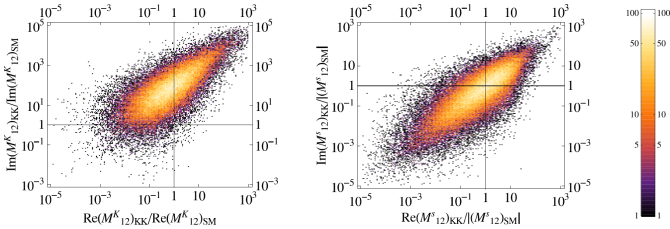
Lowest existence point, S1 and S2. The lowest M_{KK} at which *any* point passes ε_K is below the 1 TeV grid bottom for *both* scenarios: ε_K is fully tunable, so even anarchic S1 occasionally throws an accidentally-aligned point (small $\text{Im } M_{12}$) that is consistent at 1 TeV. What S1 $\rightarrow$ S2 changes is the *density* of passing points, not this floor: at $M_{KK} \leq 3$ TeV, S1 has only ~ 470 consistent draws (a fluke), S2 has $\sim 21,000$ ($\sim 44\times$ more, the $U(3)$ alignment makes consistency generic). So alignment lowers the *typical* floor (~ 10 to $\sim 2\text{--}3$ TeV), not the existence floor (< 1 TeV for both).

2.5 Consistency fraction, S1 (Bauer et al. 0912.1625, Fig. 5)



Fraction of **S1** (standard) anarctic points consistent with $|\epsilon_K|$ vs M_{KK} , the black solid curve in Bauer's Fig. 5; the $1/M_{KK}^2$ decoupling matches the paper.

2.6 $\Delta F = 2$ amplitudes, S1 (Blanke et al. 0809.1073, Fig. 2)



Re vs Im of the KK contribution to M_{12} for K and B_s (**S1** ensemble). For kaons the imaginary part dominates the real part by $\sim 100\times$; ours is $106\times$.

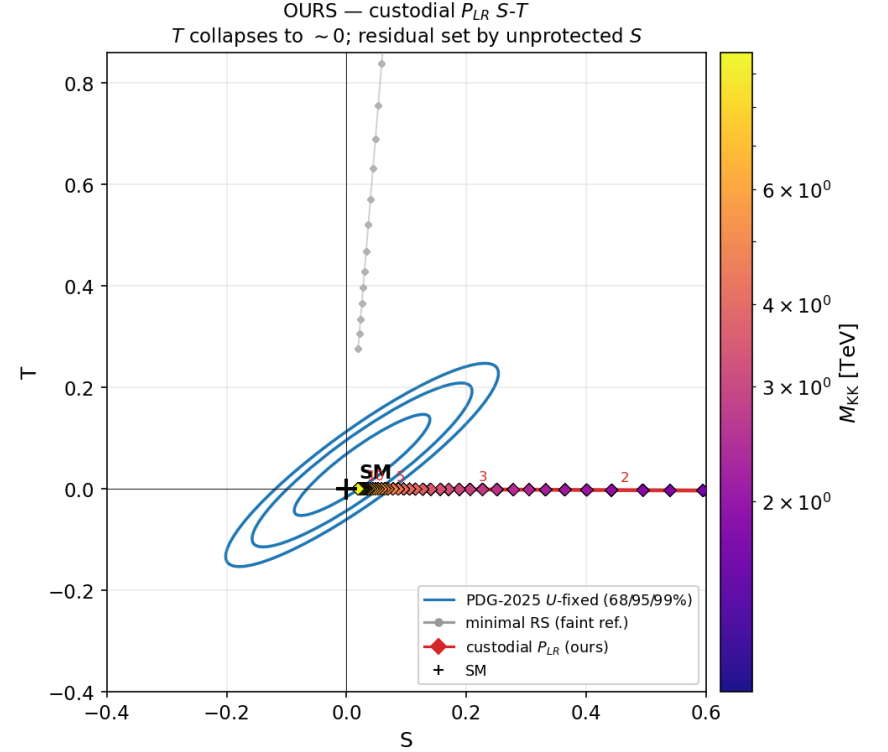
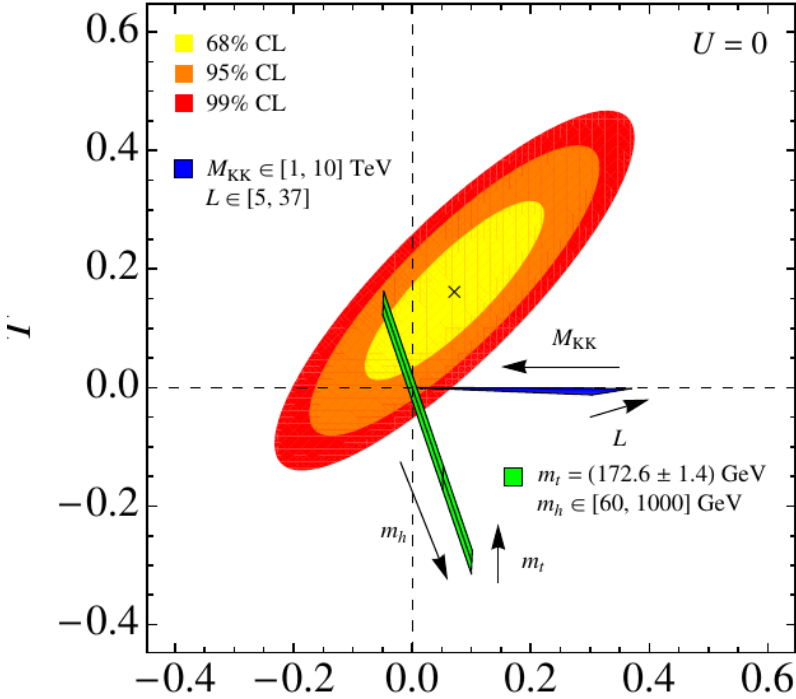
3 Custodial symmetry: what it changes

Custodial RS ($SU(2)_L \times SU(2)_R \times U(1)_X \times P_{LR}$) is an **electroweak** cure. $SU(2)_R$ protects the oblique T ; the discrete P_{LR} protects the Zb_L vertex. It does **nothing** to ε_K , because that is set by the color-octet KK-gluon C_4 , an electroweak singlet. Only flavor alignment moves the ε_K wall. The two protections are separable (oblique floor vs PDG-2025 U -fixed ellipse; δg_L^b at $M_{KK} = 3$ TeV):

EW model	oblique S, T, U floor	Zb_L shift δg_L^b
minimal RS	16.0 TeV (T problem)	2.6×10^{-3} (unprotected)
$SU(2)_R$ only	5.8 TeV (T fixed; set by S)	2.6×10^{-3} (= minimal)
$SU(2)_R + P_{LR}$	5.8 TeV (unchanged)	0 (protected)

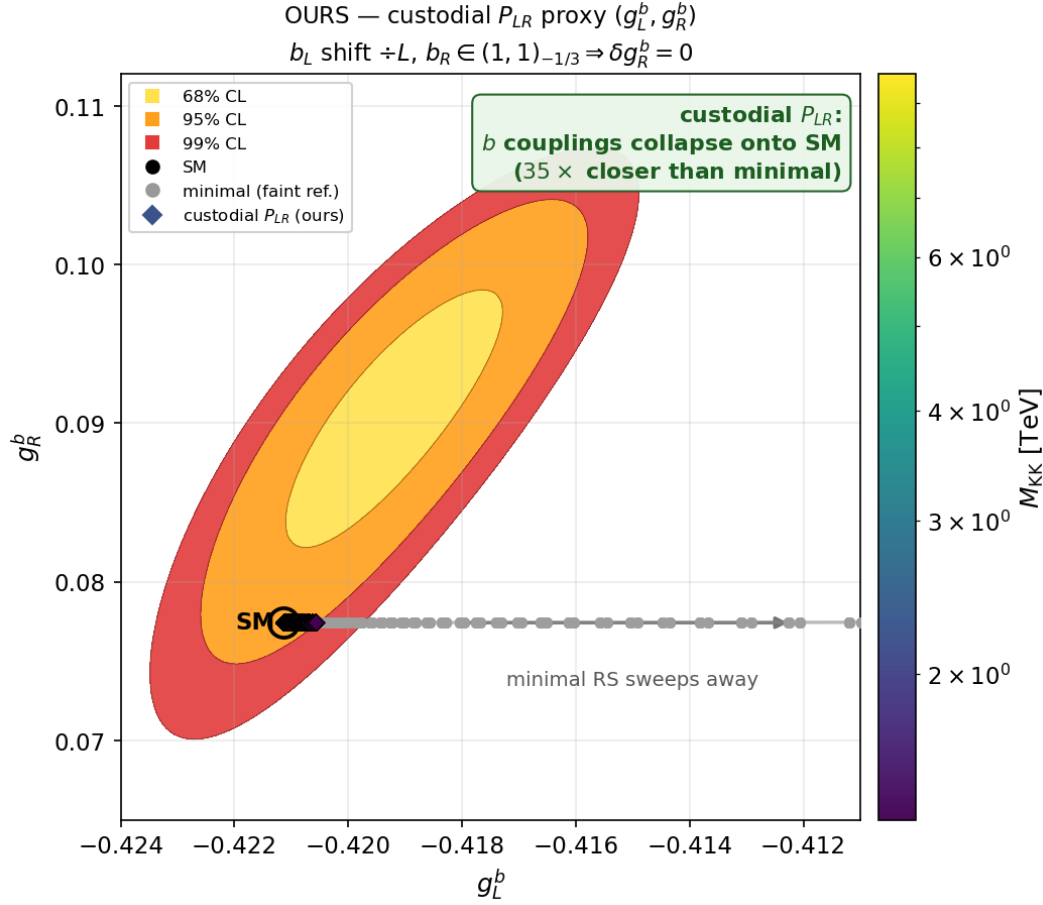
$SU(2)_R$ does the whole oblique job ($16 \rightarrow 5.8$ TeV); P_{LR} collapses δg_L^b without touching the oblique floor. Independent cures.

3.1 S, T, U with custodial (CGHNP 0807.4937, Fig. 4 right panel | ours)



The custodial T collapses to ≈ 0 (the volume-enhanced T is gone); the residual floor is the *unprotected* S . Left is the literal right half of the same Fig. 4 whose left (minimal) half is the S, T, U reproduction above.

3.2 $Z \rightarrow b\bar{b}$ with P_{LR}



P_{LR} collapses the b couplings onto the SM (δg_L^b suppressed by $1/L$, $\delta g_R^b \rightarrow 0$ for $b_R \in (1, 1)_{-1/3}$), $\sim 35\times$ closer than the minimal stripe, which sweeps away. No published (g_L^b, g_R^b) custodial panel exists to compare against.

3.3 ε_K with custodial: unchanged

Custodial is blind to ε_K : the cloud is identical to the S1 panel above, because the binding operator is the color-octet, EW-singlet KK gluon. Blanke 0809.1073 computes the full $\Delta F = 2$ inside the custodial model and still finds $M_{KK} \gtrsim 18\text{--}30$ TeV. Custodial is an electroweak cure, not a flavor cure.